African Tax Reform

AI & big data · informal sector

Regional context

Region: Sub-Saharan Africa

Case study: Ghana · vs Kenya, SA

Survey: All three surveys

Respondents: 53 · 21 · 1

1 2 3 4 5 6 7

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3 Revenue authorities compared

53 Informal workers (Ghana)

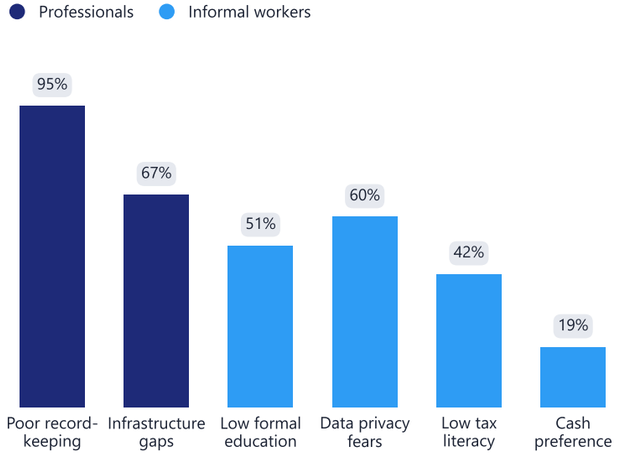
21 Tax & finance professionals

1 Telecom operator (MTN Ghana)

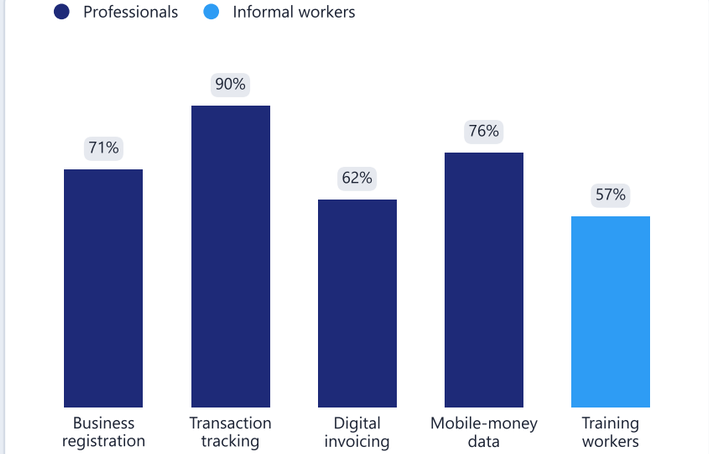
5 Policy interventions

3 Phases in the roadmap

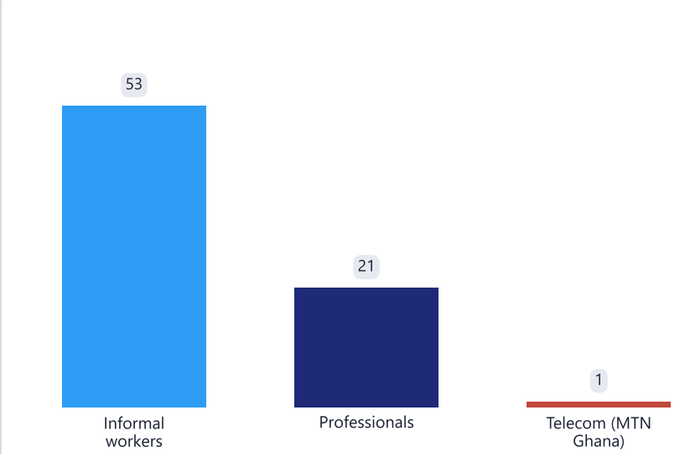
Africa's informal-sector challenges: Ghana evidence (%)



Support for the paper's key interventions (%)



Case-study evidence base (respondents)



AI in tax administration: South Africa, Kenya, Ghana

	South Africa (SARS)	Kenya (KRA)	Ghana (GRA)
Digital filing & auto-assessment	In use	In use	In use
Electronic invoicing	Not reported	In use	Not reported
ML evasion detection	In use	In use	Not reported
Mobile-money transaction data	Not reported	In use	Emerging
AI taxpayer assistant	In use	In use	Not reported
National AI / digital strategy	In use	Emerging	Emerging

Phased roadmap for African tax authorities

Short term	Pilot AI-assisted audits; digital tax literacy through trade associations, MoMo agents and radio
Medium term	Automated assessment and taxpayer profiling; e-invoicing and mandatory digital business registration
Long term	Predictive compliance models under an AI ethics committee with regular bias audits
Across the region	National ID–TIN–mobile-money linkage, learning from KRA's eTIMS and SARS auto-assessment

Sources: comparative review of SARS, KRA and GRA in the GCA paper (capabilities coded from its text); Ghana case-study surveys. Survey shares describe the Ghana sample: evidence for Sub-Saharan Africa, not a measure of it.

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Worker profile

Region: Sub-Saharan Africa

Case study: Ghana

Survey: Informal workers

Respondents: All (n = 53)

53 Ghana workers surveyed

51% No formal education

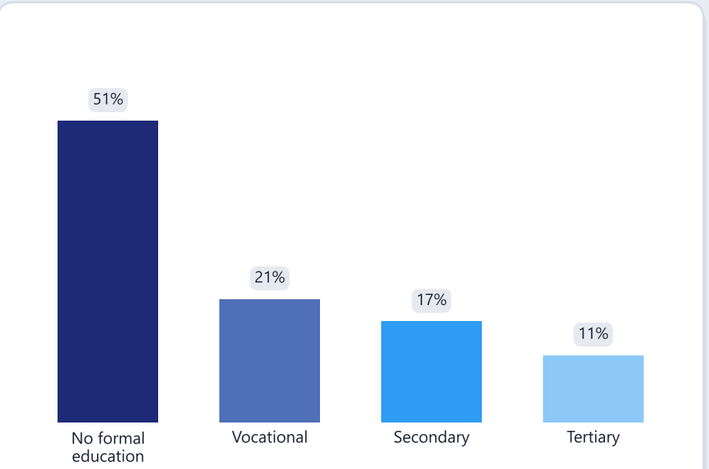
71% Earn over GH₵1,000 a month

10 yrs Median experience

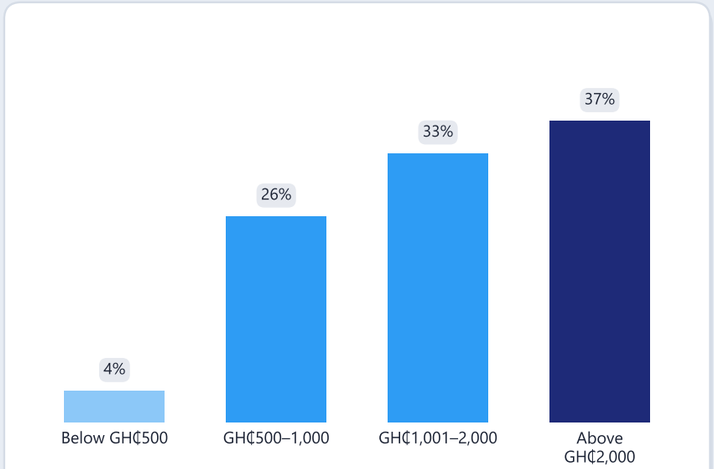
4 Types of work

51 Written comments on AI

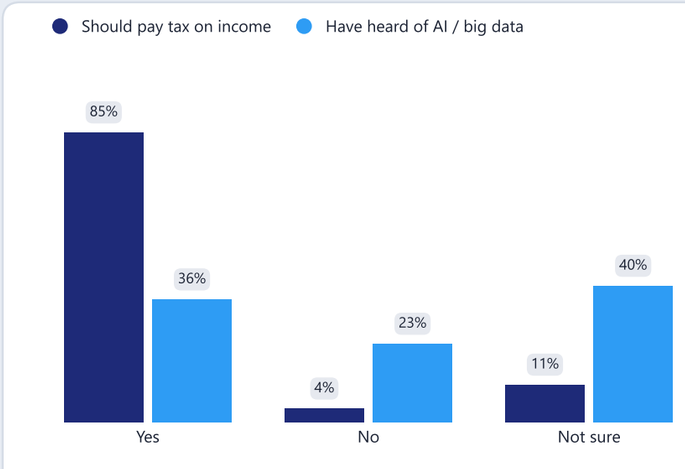
Education level (% of workers)



Monthly income (% of 51 answering)



Understanding taxes vs AI awareness (%)



How often each digital service's users go digital (% of users)

	Daily	Weekly	Monthly	Occasionally	Never
Mobile money	41%	10%	14%	33%	2%
Utility bills	44%	8%	20%	24%	4%
Market transactions	39%	13%	17%	30%	0%
Social media for business	20%	20%	20%	40%	0%

Words workers used about AI in tax collection



Largest word used 8 times. Words echoing the question (AI, big data, tax, collection) left out.

Source: Ghana case-study survey of informal-sector workers (n = 53), hand-coded from the scanned questionnaires; shares use those answering each question.

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Digital readiness

Region: Sub-Saharan Africa

Case study: Ghana

Survey: Informal workers

Respondents: All (n = 53)

98% Use mobile money

42% Use digital services daily

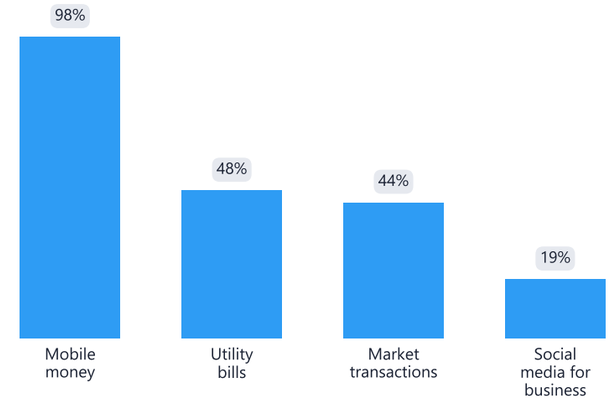
54% Say data would help GRA

44% Would share data with GRA

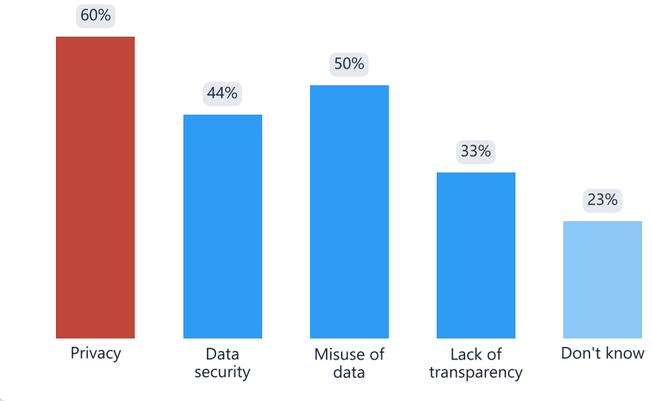
60% Worry about privacy

19% Prefer cash to MoMo

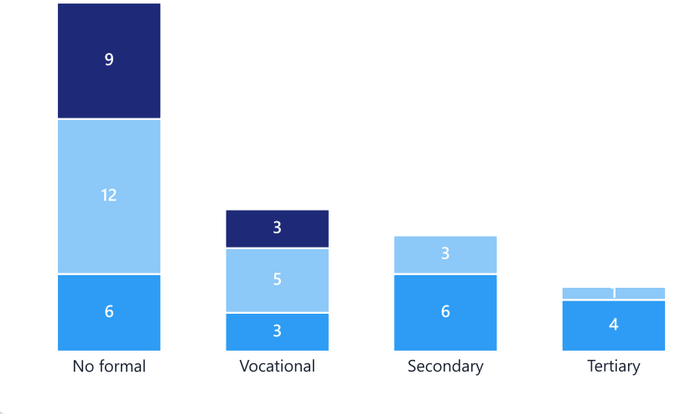
Digital services used (% of workers)



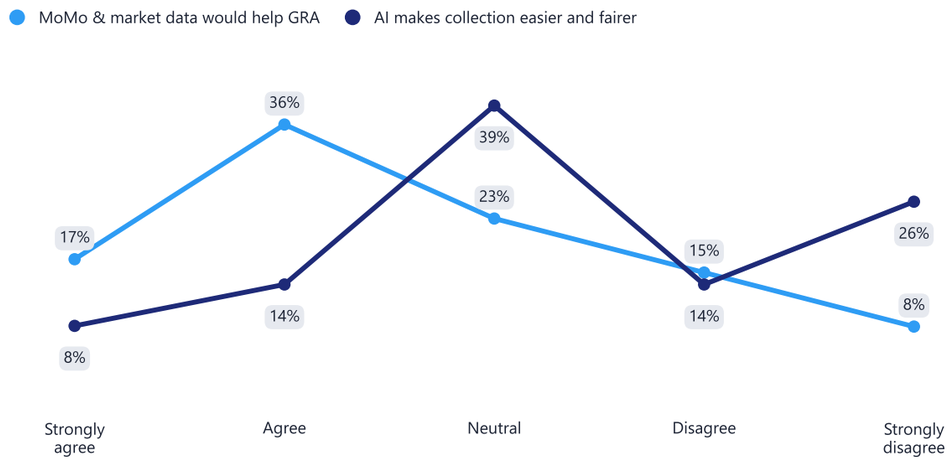
Concerns about tax use of their data (%)



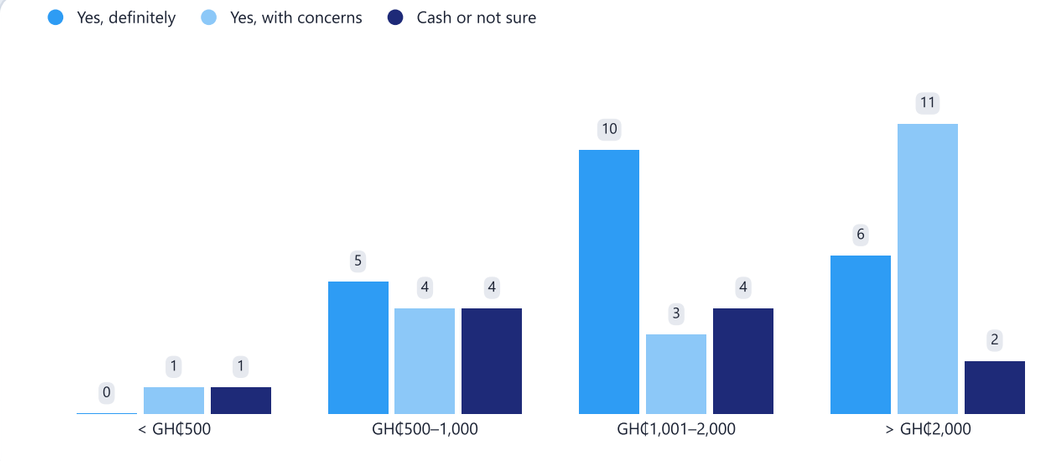
AI awareness by education (workers)



Agreement profile: MoMo data helpful vs AI helpful (%)



Comfort with mobile-money payment by monthly income (workers)



Source: Ghana case-study survey of informal-sector workers (n = 53), hand-coded from the scanned questionnaires; shares use those answering each question.

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Tax attitudes & AI

Region: Sub-Saharan Africa

Case study: Ghana

Survey: Informal workers

Respondents: All (n = 53)

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85% Say they should pay tax

76% Doubt it makes a difference

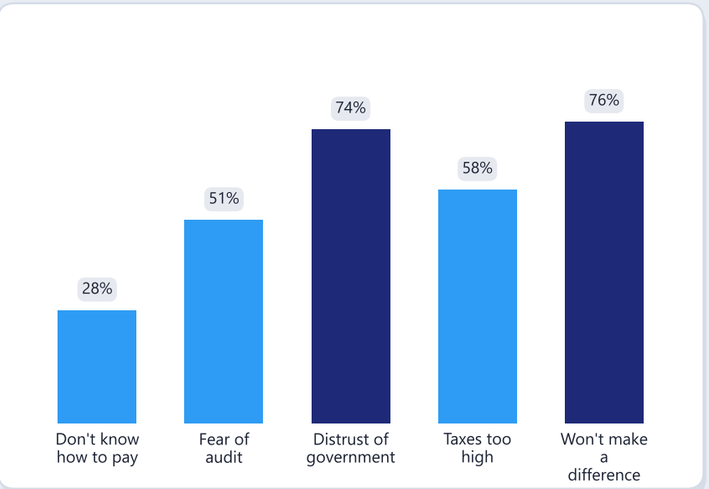
74% Distrust government

42% Can't name a tax benefit

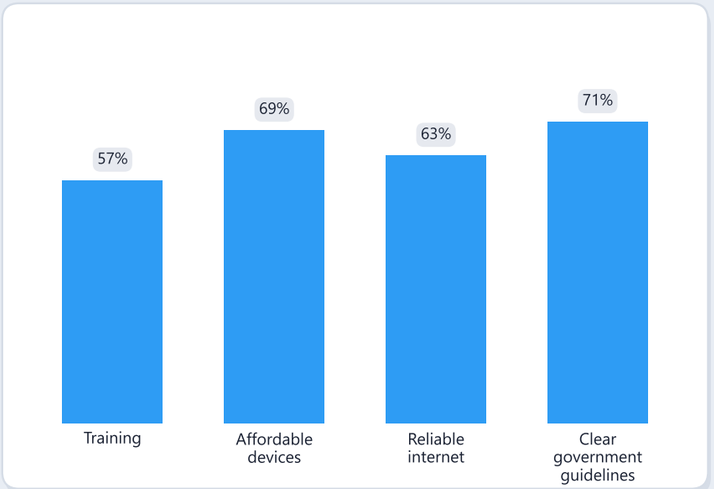
36% Have heard of AI

22% Think AI would help

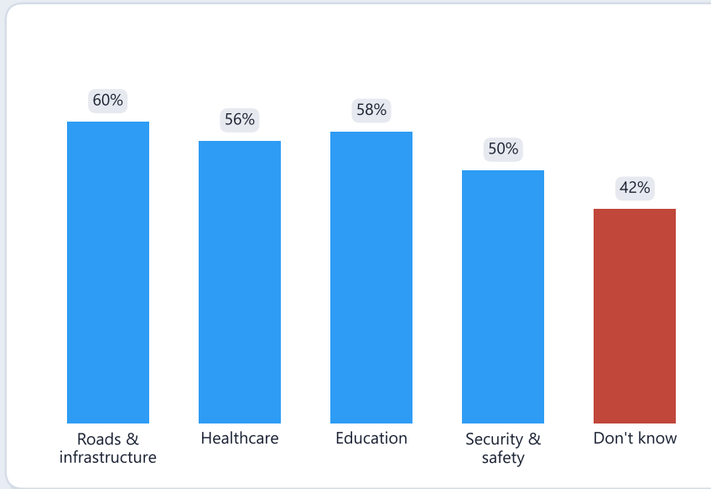
Why workers don't pay tax now (%)



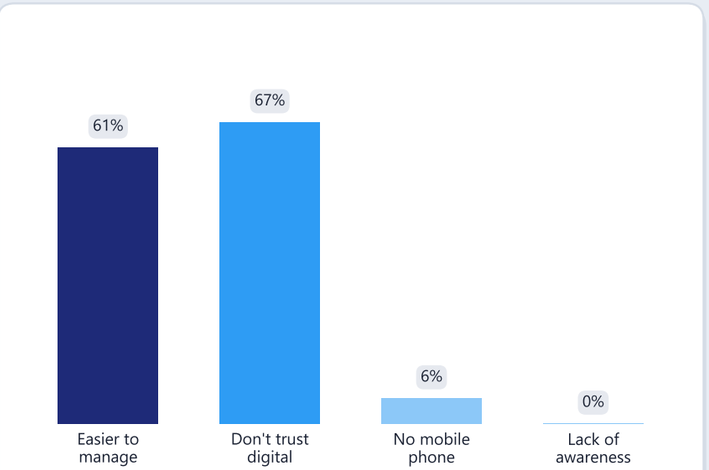
Support needed to adopt digital tax (%)



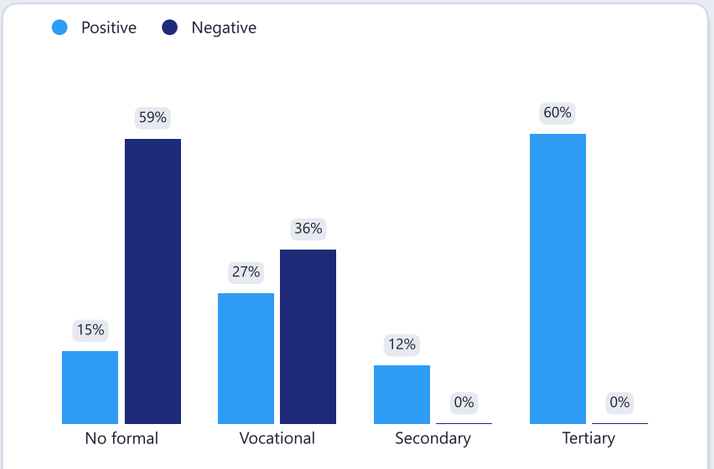
Perceived benefits of paying tax (%)



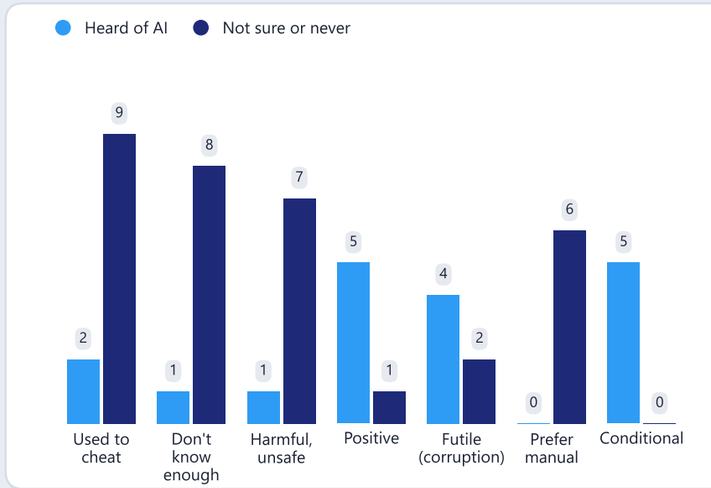
Why some prefer cash (18 answering, %)



View of AI by education (% of each group)



Comment themes by AI awareness (comments)



Source: Ghana case-study survey of informal-sector workers (n = 53), hand-coded from the scanned questionnaires; shares use those answering each question.

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Readiness segments

Region: Sub-Saharan Africa

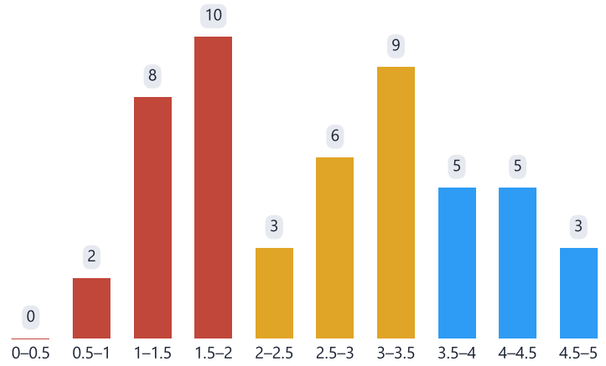
Case study: Ghana

Survey: Informal workers

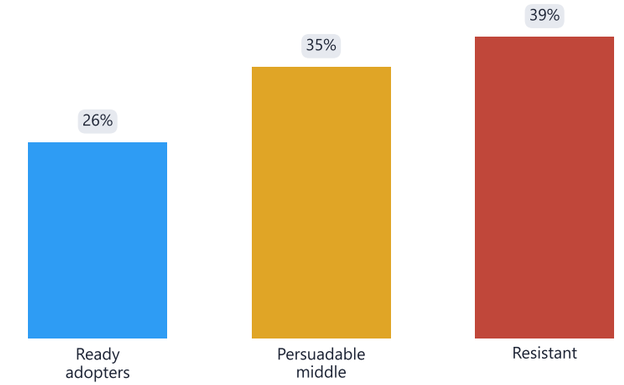
Respondents: 53 (51 scored)

26%	35%	39%	56%	3%	-0.07
Ready adopters	Persuadable middle	Resistant	Positive if aware of AI	Positive if not aware	Income × AI view (ρ)

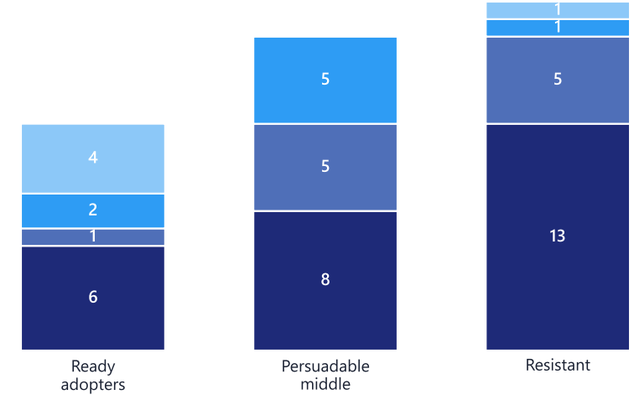
Readiness score distribution (workers)



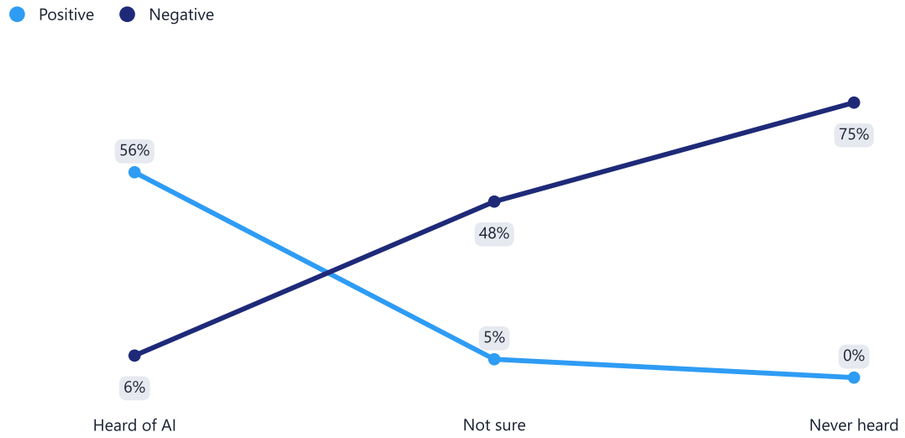
Segment size (% of workers scored)



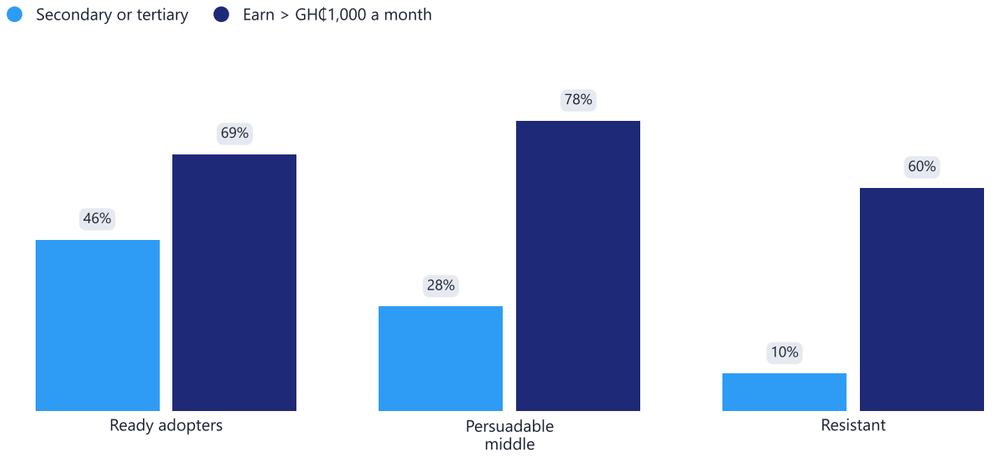
Segments by education (workers)



Awareness shifts opinion: positive vs negative view of AI (%)



Segment profile: education and income (% of segment)



Source: Ghana case-study survey of informal-sector workers (n = 53), hand-coded from the scanned questionnaires; shares use those answering each question. Readiness score 0–5: digital use, MoMo comfort, data sharing, AI awareness, AI view; cut-offs 2.25 and 3.5.

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Professionals

Region: Sub-Saharan Africa

Case study: Ghana

Survey: Professionals

Respondents: All (n = 21)

1 2 3 4 5 6 7

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21 Professionals surveyed

67% Call today's system ineffective

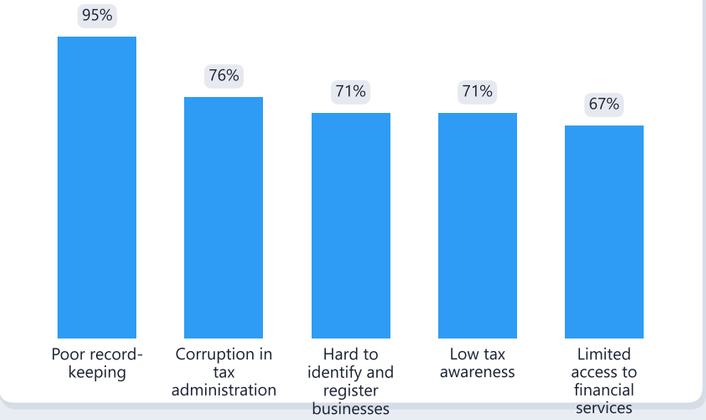
95% Cite poor record-keeping

86% See privacy as top AI risk

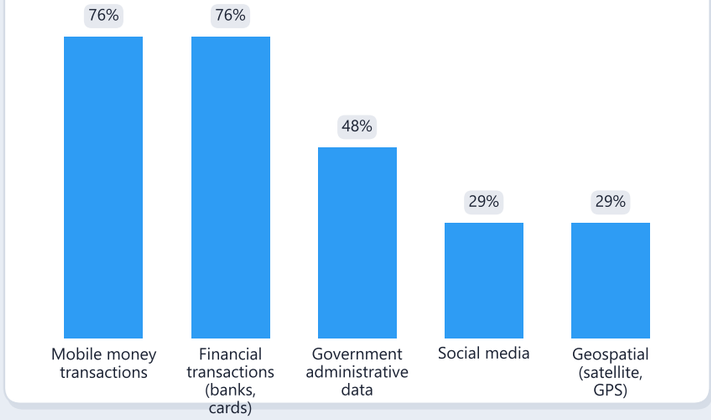
57% AI tax feasible without big investment

57% Expect a broader base first

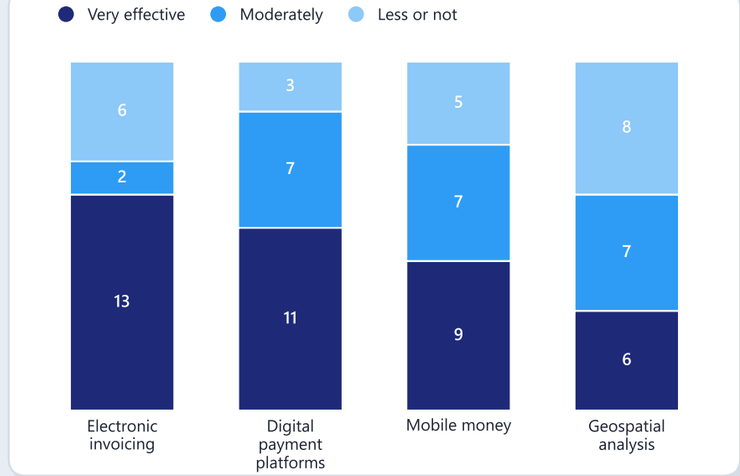
Top collection challenges (% of professionals)



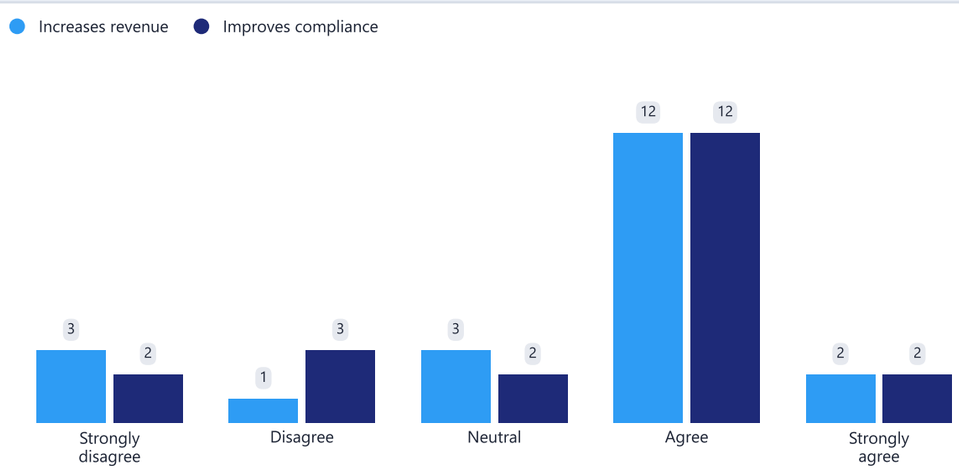
Most crucial data sources for AI (%)



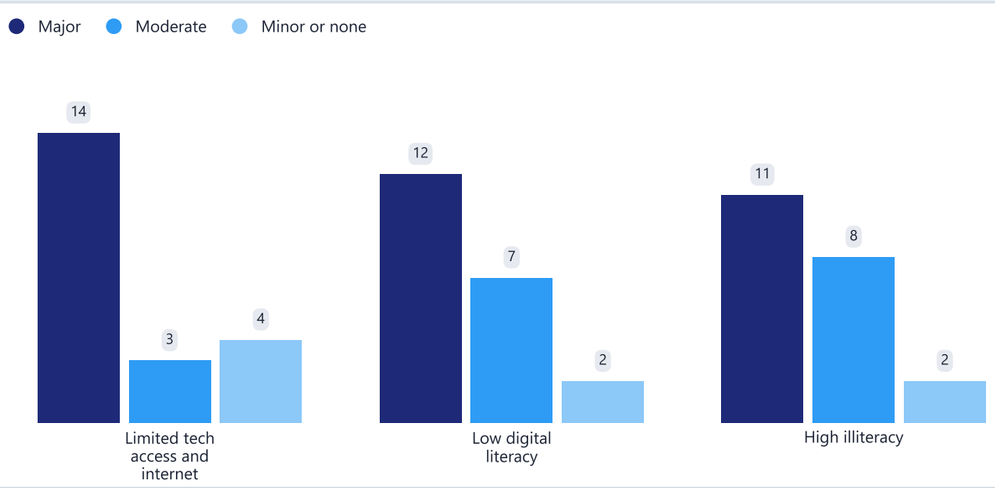
Effectiveness of existing technologies (professionals)



Would limiting cash help? (professionals)



Obstacles to AI in informal-sector taxation (professionals)



Source: Ghana case study, professional survey of accountants, auditors, tax officers, economists and data analysts (n = 21), Google Forms summary.

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Validation & MTN

Region: Sub-Saharan Africa

Case study: Ghana

Survey: All three surveys

Respondents: 53 · 21 · 1

1 2 3 4 5 6 7

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27/29 — Paper figures within 5 pts

-17 — Secondary-education gap (pts)

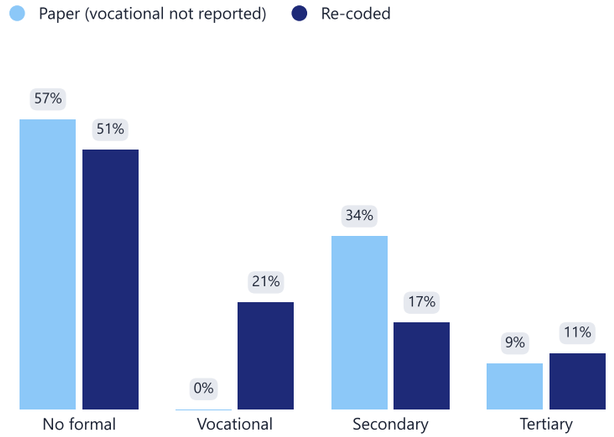
21% — Vocational, not in paper

1 — MTN Ghana response

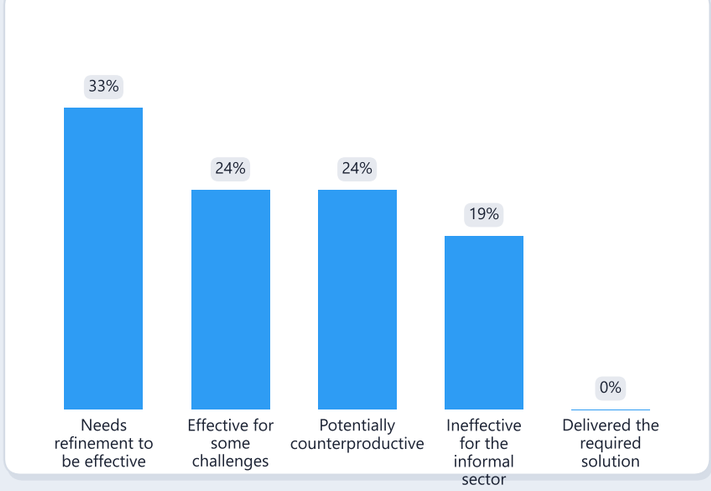
33% — Say E-Levy needs refinement

38% — Pick transparency as lever

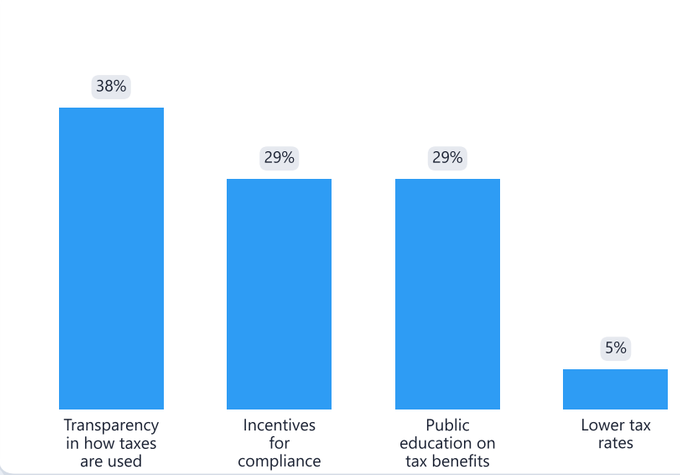
Education: paper vs re-coded forms (%)



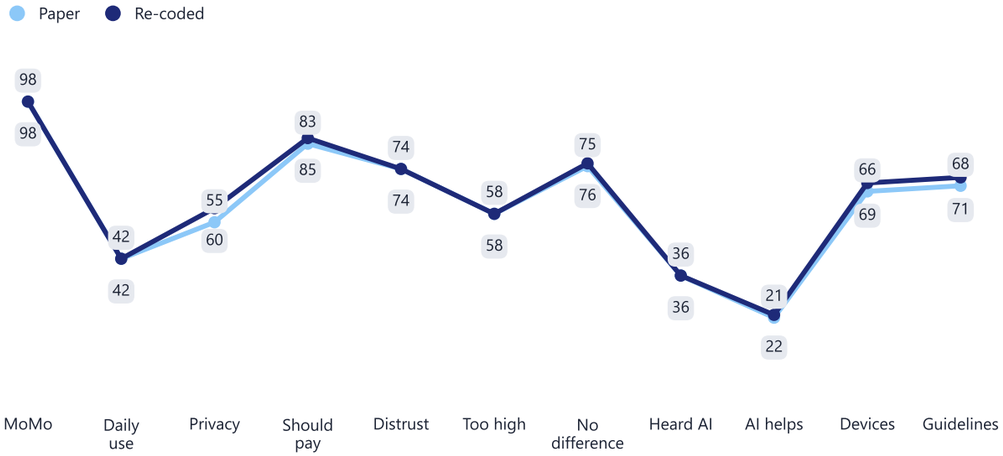
Professionals' verdict on the E-Levy (%)



What would most encourage compliance (%)



Headline figures: paper vs re-coded data (%)



MTN Ghana: ready to share data, not to run the tax tool

Uses AI for	customer analytics, SIM-swap fraud detection, revenue management
Would share	mobile-money transaction data with GRA
On condition of	legal frameworks, data security and clear privacy rules
Worker tax tool?	No: outside MTN's scope; GRA must own it
Proposes	automatic tax deduction from MoMo transactions
Expects	workers to resist digital platforms for fear of tax

Sources: informal survey (n = 53, re-coded) compared with the GCA paper draft; professional survey (n = 21); MTN Ghana questionnaire (one institutional response).